Consulting Report
: Maximizing Profitability through Vertical Integration and Strategic Alliance

Team 36
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Executive Summary

1. Context:

Post-crisis in aircraft reliability, Boeing's **BCA division** continues to lose share and customer confidence to Airbus.
A **recovery strategy** is required to **rebuild trust, stabilize performance, and reposition for sustainable growth**.

2. Challenge:

- **High supplier dependency** (>$34B spend) driving high cost, parts delay, and quality issues.
- **Intense rivalry** within duopoly limiting market share/revenue growth.

3. Strategic Recommendations

A. Rivalry Management

- Explore ***strategic cooperation*** *with Airbus* on mutually-reinforcing initiatives to improve market growth.
- Strengthen differentiation through Boeing's core advantages in **Integration and Reli**ability by focusing on wide-body aircraft and cross-family commonality.

B. Supplier Management

- ***Vertically integrate*** **Tier-1** aerostructures (e.g., Spirit)
- *Closely monitor/partner with* selective Tier-2/3 suppliers.

4. Expected Impact

- Annual **+5.5%** in Revenue, **+1.2%** market share/Year (~2030).

Industry Analysis

: The BCA market, being more susceptible to market competition, whereas the BDS market operates in a more controlled environment, shows greater scope for improvement through strategic initiatives.

Defense, Space and Security (BDS)

Market Share (Competitors)
Lockheed(15%), RTX (6.1%), Boeing(4.8%)
Internal Revenue Percentage
36% of Total Revenue ($23.92B)
Market Growth Expectations
+17.9%/Y (Industry Norm: 14.5%)
Market / Customer Analysis
Market Size: Bigger market cap compared to BCA **Sole Customer**: US Defense budget/contracts. **Competition**: moderate, highly controlled by government

Stable market with guaranteed sustainable growth.
Less autonomy for firms
(Controlled by powerful customer)

Commercial Airplanes (BCA)

Market Share (Competitors)
Airbus(56%), Boeing(40.%) → **Duopoly** (2024).
Internal Revenue Percentage
34% of Total Revenue ($22.86B)
Market Growth Expectations
Passenger Traffic **+4.2%/Y** (for 20 Years)
Market / Customer Analysis
Market Size: Smaller compared to BDS **Customer**: Worldwide commercial airlines. **Competition**: High competition btw Duopoly (Unlikely to steal market % from Airbus)

High competition (duopoly), constantly **losing market share to** bigger competitor (Airbus).
High impact after Boeing quality management crisis.

Industry Analysis – 5 Forces / Diagnosis

: With high rivalry and supplier power posing the greatest threats to the BCA industry, Boeing must prioritize risk mitigation strategies focused on **1.Rivalry management** and **2. Supplier management**.

Buyer Power (Mid)

- Airlines are price sensitive and demand highly customized aircrafts, limiting comparability within providers.
- A lot of customers are on backlogs and long-term contracts
- High switching costs as involved with pilot retraining, maintenance etc.
- Small-medium sized airlines purchasing a few, whereas only mega-buyers own negotiating power (Emirates, Ryan Air)

Threat of New Entrants (Low)

- Duopoly (96%) threatening new entrants through economy of scale, rapid innovation. Highly likely to attack new players through price reductions, leveraging existing customer loyalty, lobbying, likely to end in absorption. (eg. Bombardier & Airbus)
- High investment costs for aircraft development ($10-15B), sunk if project failure.
- Regulatory barriers(FAA/EASA) limiting new entrants through excessive and time-consuming testing, compliance requirements.

Rivalry (High, but controlled)

- Duopoly: Major competition btw Boeing and Airbus, with minor pressure from other players.
- Innovation rivalry: Competition centering on fuel efficiency, cost-saving, and digital tech, driving differentiation.
- Market Stability: High-cost structures and long product cycles encouraging strategic rivalry.

Threat of Substitutes (Low)

- Used and leased airplanes are getting more prevalent, reducing demands for new airplanes.

Supplier Power (High)

- For parts of low importance(commodity, low-tech), substitutability is high.
- However, with parts that are more related to safety or requires high-tech(avionic modules, cabin interiors), substitutability is lower.
- Parts/modules that are highly essential for aircraft completeness and safety(engines, primary flight modules), supplier power is greatly high due to small number of suppliers, high requirements of technology (from firms, government) and high switching costs.

Company Analysis – Rivalry Management

: Airbus's differentiation and lead comes from Efficiency and Innovation**. Boeing** needs to strengthen its differentiation strategy on **Integration and Reliability,** utilizing its existing customer base and resources.

Dimensions	Boeing	Airbus
Strategy	**Integration**: emphasis on system integration, vertical integration for steady ecosystem. **Reliability**: Focus on existing customer relationships for repeat orders, focus on stabilization and QA recovery	**Efficiency**: common platform designs (A320 line) and manufacturing efficiency(digital integration) **Innovation**: Steady launch of new product line (A320neo, A321XLR, A350)
Objectives	1. Restoring **production stability** and regaining customer trust. (Damage control after QA issues) 2. Achieve **gradual improvements** (e.g., 777X, lightweight materials) than radical innovations to mitigate risk.	1. Expanding leadership on **narrow-body planes** (A320neo, A321XLR) 2. Focus on **sustainable technology and digital services** as differentiation (e.g., hydrogen-powered/ electric hybrid initiatives).
Assumptions	Sustainable sales come from **customer relationship management** than innovational fleets. Airlines(customers) value **operational reliability, aircraft safety/quality** with long-life-cycle, high-cost products.	**Standardization** of fleet is key for sustainable sales. Airlines(customers) will increasingly prioritize **environmental impact** and **fuel efficiency**.
Resources & Capabilities	**Strength**: Large existing customer base (high switching costs for airlines, probable to be loyal customers) **Weakness**: Production & quality reliability issues, slow innovation cycle with limited ongoing projects.	**Strength**: Efficient manufacturing cycle with unified system/design throughout product line. (Lower airline training/maintenance costs) **Weakness**: Less diversified product portfolio as well as expenditure on long-term projects(sustainability).

Company Analysis – Rivalry Management

: Boeing's sustained advantage lies in its **wide-body dominance, integrated fleet design, data-driven operational depth, and lifecycle ecosystem** that together create durable customer loyalty.

Value : Wide-body aircraft dominance & long-range operational reliability *Boeing* serves to be the world's largest manufacturer for long-distance wide-body jets (777,787 family with 1.8K+, 1.1K+ orders cumulative) *Airbus*: A350/A330neo are competitors that lack 777's range and capacity economics.	**Rare : Cross-family operation logic** *Boeing* provides similar cockpit logic/design to allow pilots to easily operate on cross-family flights (narrow vs wide body) *Airbus*: Achieves cockpit commonality within same families (within narrow/wide body) but lacks integration across.
Inimitability : Cumulated Operation data collection *Boeing,* sourcing 10K+ aircrafts with a cumulative 1.3B flight hour data is available to source data-driven insights (eg. airplane health/maintenance prediction). *Airbus*: With a shorter corporate history and depth, Airbus lacks to achieve the same level of analytical resources.	**Non-substitutability : Lifecycle management** *Boeing* provides maintenance, pilot training, binding customers long-term, and once accustomed, threatens to be high in switching costs (unlikely to learn new ecosystems from scratch). *Airbus*: Less emphasis on aftersales market and smaller training / customer loyalty ecosystem.

Company Recommendation - Rivalry Management

: Pursue industry **collaboration for sustainable market growth** while reinforcing Boeing's wide-body leadership and integration strengths to improve competitiveness in the optimized market.

Strategy	**1. Form Industry Partnerships for Sustainable Growth**	**2. Concentrate on Wide-Body Leadership & Cross-Family Integration**
Objective	Introduce **co-operative measures** to promote **long-term, sustainable market growth** while managing heated competition.	**Consolidate dominance** in high-margin market-leading products(wide body 777, 787) and **leverage synergy** across product lines.
Details	- Share non-competitive initiatives to improve **aviation sustainability** (eg. green energy development) . - Leverage joint advocacy to shape mutually beneficial **environmental policy.**	- Focus R&D investment on **wide-body airflight innovation**. (eg. 777X – projected for 20% of total BCA Revenue) - Enhance **cross-family integration** to strengthen brand competitiveness by lowering airline operating costs. (eg. pilot training).
Reason	Joint efforts would **reduce redundant R&D spending,** mitigate risk as the less-innovative player while **optimizing the market overall.**	Focusing on products/technology that aligns with Boeing's strengths in **Integration and Reliability** will boost market appeal through operational stability and customer trust.

Company Analysis – Supplier Management

: Approximately $34 Billion is spent on suppliers, which vary by **safety, technology, and supplier concentration,** of which most is spent on **high-importance(Tier 1)** components.

Overall Supply Chain Costs:

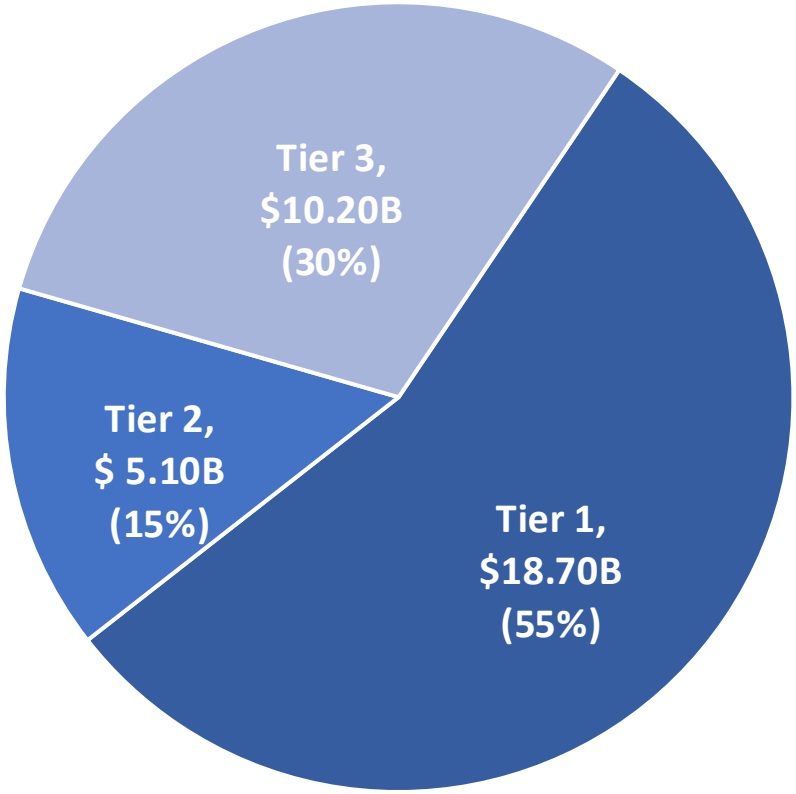
- COGS: 90% of Revenue ($70.07B)
- Network of 11K+ global suppliers
- **$34B+ expenditure** in parts procurement (1B+)

➔ Heavy cost structure : may indicate possibility for **cost reduction through integration/process innovation.**

Supplier Significance Categorization

- **Tier 1** : Critical to aircraft safety & requires high technology.
- **Tier 2** : Parts of high importance to aircraft structure, mid tech requirements.
- **Tier 3** : Interior / passenger-experience related parts, low tech, high room for substitutability

Estimate of Expenditure per Significance



Company Analysis – Supplier Management

: Tier 1 suppliers face the highest coordination and contracting challenges, coupled with advanced technology and critical impacts on the final products, making them strong candidates for integration.

Supply Chain Corporate Scope Analysis

- **Tier 1** : ↑ Coordination, ↑ Contracting Difficulty

Coordination: Critical, high value systems with complex integration needs within parts, limited supplier

Contracting: High tech requirements, limited supplier base

- **Tier 2** : ↑ Coordination, Moderate Contracting Difficulty

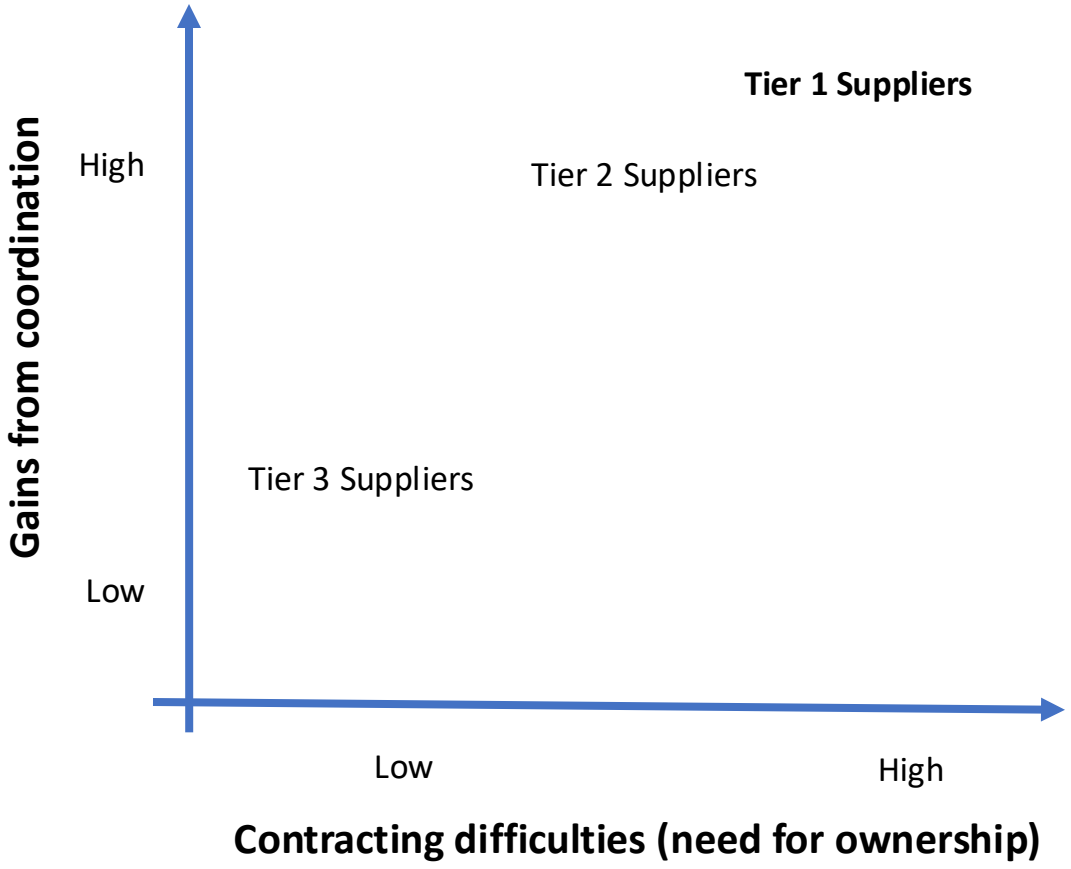
Coordination: Specialized components requiring same control logic for coordination.

Contracting: Moderate tech requirements, less-limited supplier base.

- **Tier 3** : ↓ Coordination, ↓ Contracting Difficulty

Coordination: Standardized, low-value commodities.

Contracting: Abundance of suppliers, high bargaining power(Boeing).



Company Analysis – Supplier Management: Tier 1

: Amid recurring financial losses due to **supplier delays and quality issues**, Boeing's successful Tier-1 acquisition of Spirit highlights the tangible **benefits of stronger supply chain oversight through integration.**

Case evidence where impact by underperformance of Tier 1 Suppliers

Case	Issue	Impact	Consequences
737 MAX (2019-21)	Sensor& system integration failure	Entire fleet grounded(20months), halted deliveries, lost production rate	**- $30B** (compensation, cancelled orders, direct costs)
777X (2024)	Engine-to-body part failure	Certification schedule pushed 9-12 months, delay in deliveries	**-$1B** (lost margin, penalty), reputational risk
787 (2023-24)	Supplier misalignment on specs / quality	Delivery delay, production bottleneck.	**-$10B** (deferred costs due to delays)

Case precedent of Boeing's Selective Integration decreasing costs : Spirit AeroSystems

2005

Spirit inherits Boeing's structural work(fuselage, wings) as a Tier 1 Supplier

2023

Increasing discontent of Spirit parts' quality, delays with frequent reinspection/bottleneck

2024 July

Boeing acquires Spirit for $8.3B ($4.7B Equity)

Current

Projected **$1B/Y synergy** by 2026 from reduced overhead, streamlined procurement

Expected **margin improvement of 10-15%** for integrated operations

Company Recommendation – Supplier Management: Tier 1

: Given its significance and high supplier power, **aerostructure should be fully integrated** to secure value and control cost whereas other parts/systems require less management.

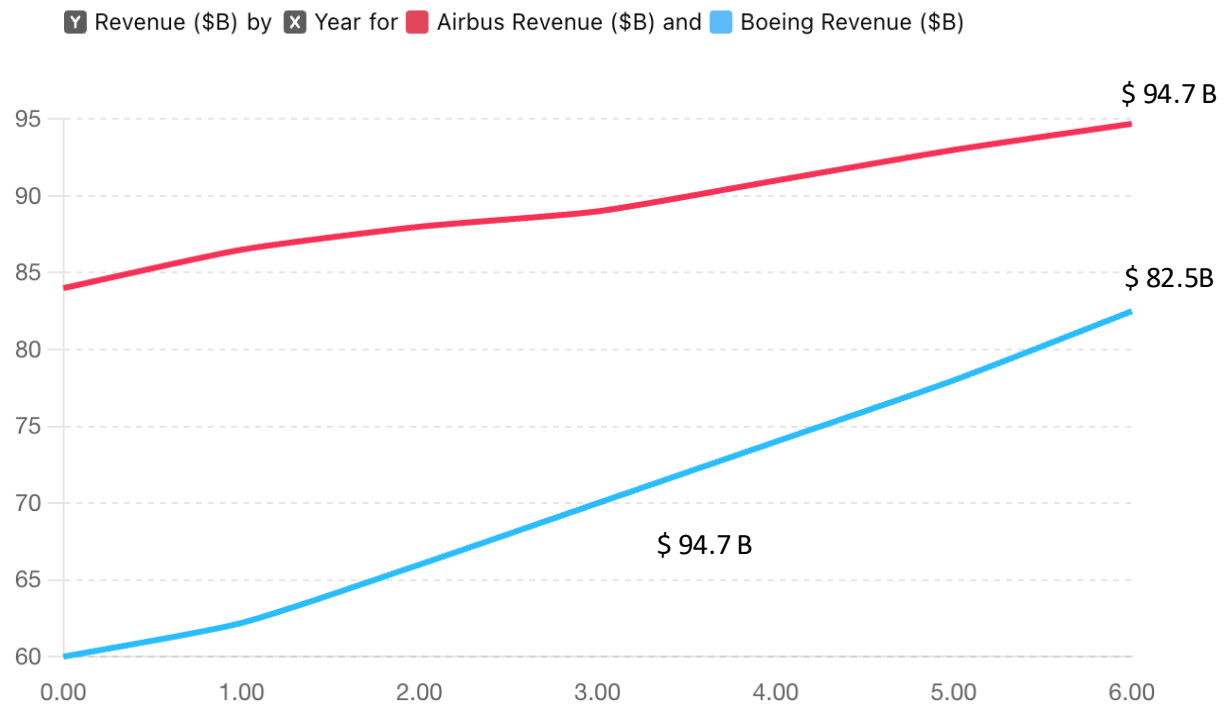
Candidate Parts for Vertical Integration

Product	Reason for Consideration	Feasibility	Potential Benefits	Strategy
Aerostructure (Fuselage, Wings, Nacelles)	• Highest spend/importance **(45% of T1 budget)** • Direct link to aircraft safety and final product reliability • Past quality issues causing manufacturing backlog.	**High**	**$1-1.4B/Y** savings via elimination of supplier margins **2-3%/Y production growth** through faster aircraft ramp-up	**Full integration** **:** With preceding progress in Spirit, gradually consolidate composite/part procurement.
Flight Control & Avionics Systems	• Direct link to aircraft safety • Supplier fragmentation in systems (sensors, computers) creating flight-control logic confusion.	Moderate	**$0.2-0.3B/Y** savings of related costs **10-15%** reductions in QA defect	**Monitor** **:** Focus on software QA & integration authority, not full takeover
Engines & supporting structures	• Direct link to aircraft safety, may lead to costly program delays • Low count of suppliers (high supplier power, switching costs)	Moderate-Low	**$0.3-0.4B/Y** savings of related costs **1-2% CAGR** tech value if joint tech ventures	**Monitor/Partner** **:** Focus on QA control, co-development for timely, market-leading technology.

Strategy Effectiveness Forecasting

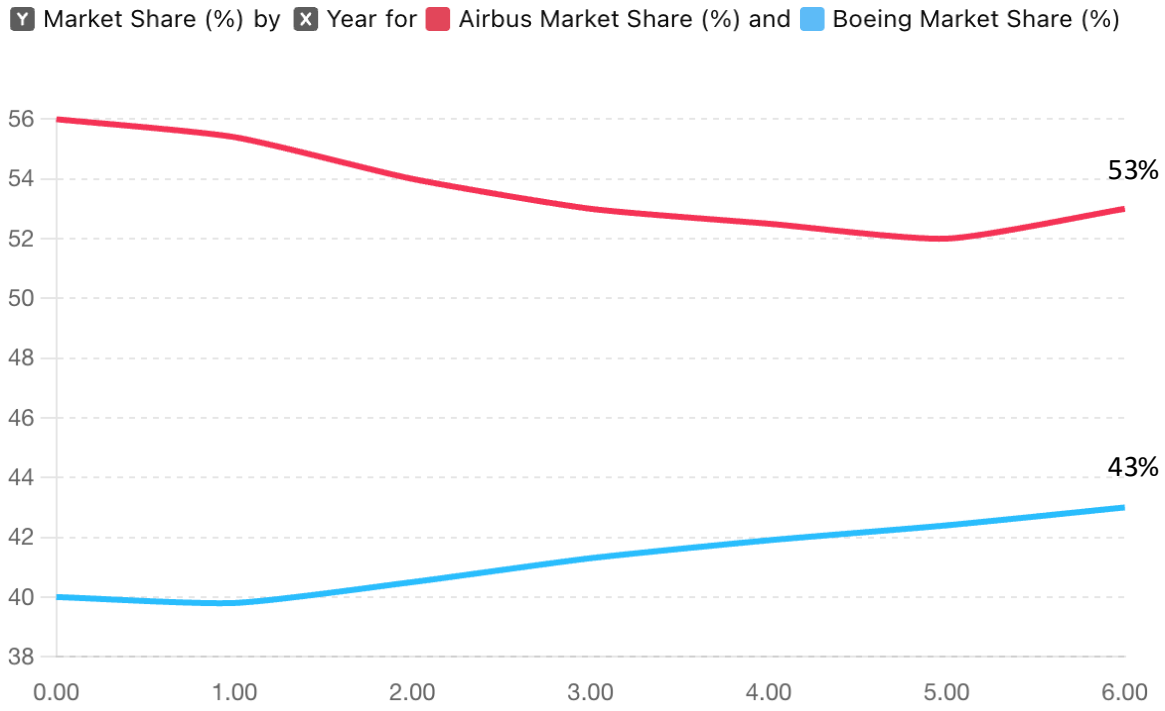
: With the two strategic recommendations enforced, Boeing is expected to gain **a 5.5% growth in revenue and 1.2% growth in market share** annually until 2030.

Revenue Forecast: Boeing vs Airbus (2024–2030)



➔ Boeing Revenue CAGR: +**5.5%** (per Year)
➔ Airbus Revenue CAGR: +2.0% (per Year)

Market Share Forecast: Boeing vs Airbus (2024–2030)



➔ Boeing Market Share CAGR: **+1.2%** (per Year)
➔ Airbus Market Share CAGR: -0.9% (per Year)

Appendix

Potential Value for Vertical Integration / Tier1 Product

Product	Potential Benefits
Aerostructure (Fuselage, Wings, Nacelles)	**$1-1.4B/Y** savings via elimination of supplier margins **2-3%/Y production growth** through faster aircraft ramp-up
Flight Control & Avionics Systems	**$0.1-0.2B/Y** savings of related costs **10-15%** reductions in QA defect
Engines & supporting structures	**$0.3-0.4B/Y** savings of related costs **1-2% CAGR** tech value if joint tech ventures

Formula

Aerostructure:

- Industry precedent: Elimination of 5–7% supplier margin ,1–3% process/QA inefficiency after integration

- Benchmark of Airbus's target: partial reintegration of Premium Aerotec targeted ~7% cost reduction.

- Boeing's 2005–2015 outsourcing issues (Spirit Aero, 787 fuselage rework) implying potential >6% efficiency recovery.

➔ $8.4B × 6–8% = $0.50–0.67B annual savings +production gains = **≈$1.0–1.4B total annual benefit**

Flight Control & Avionics Systems :

- Moderate share (~15% of Tier 1 = ~$2.8B).

- Historical examples (Boeing–Honeywell partnerships, Airbus–Thales) show 3–5% cost synergy from integration.

➔ $2.8B × 4–5% = **$110–140M**

Engines/Supporting Structures:

- 25–30% of Tier 1 (~$4.7–5.6B).

- Benchmarks (Safran–Airbus partnerships, GE–CFM collaborations) 1–3% cost improvements

➔ $5B × 2–3% = **$100–150M** + technology synergy (1-2% CAGR) **= 0.3-4B**

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LLM Usage (Prompts Used)

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